QUIZ 09, PRACTICE PROBLEMS

STAT 1110: Section – Fall 2026 with Dr. Ethan P. Marzban

⚠ Caution

These problems are not meant to be exactly indicative of what will show up on the Quiz on Monday, but are rather designed to be a starting point for your studies.

Some General Notes for this Quiz:

- The quiz will cover **both** the simulation-based approach and theory-based approaches to testing multiple proportions (sections 8.1 and 8.2).
- There will be a handful of multiple-choice and open-ended questions.
- Look at your notes on from lecture on Monday November 9; I may have already revealed one of the questions on the upcoming quiz...

Scenario 1: For Multiple Choice Questions 1 - 5 and Free-Response Questions 1 - 2.

A (fictional) study recruited 150 adult volunteers aged 18 - 28 to undergo a study that lasted for one night, designed to test the effectiveness of three different **pre-bedtime routines**: scrolling through your phone, listening to soothing music, and doing nothing (and going straight to bed). These volunteers were randomly split into three groups: one group was instructed to scroll through their phones right before bed, another group was instructed to listen to soothing music before bed, and the third group was instructed to do nothing before bed (and just try to go straight to sleep). The next morning, the volunteers self-reported **whether they felt they got a good night's sleep or not** ("good nights sleep" is a "success"). Our research question is: **is there an association between the pre-bedtime activity and whether or not you get a good night's sleep?**

Here is a summary of the data:

	Phones	Music	Nothing	Total
Good Sleep ("Success")	25	40	30	95
Not Good Sleep ("Failure")	25	10	20	55
Total	50	50	50	150
Sample Proportion	0.5	0.8	0.6	0.633

Table 1: Summary of Ayoade's data

FREE-RESPONSE QUESTION 1

Calculate the value of the mean group difference statistic.

MULTIPLE CHOICE QUESTION 1

Are the validity conditions met to conduct a theory-based χ^2 test?

(**A**) **Yes**, because we have at least 10 successes and at least 10 failures

(**B**) **Yes**, because every value in the interior of the two-way table (25, 40, 30, 25, 10, and 20) are at least 10

(**C**) **Yes**, because the group sizes (50, 50, and 50) are all greater than 20

(**D**) **No**, because even though the group sizes (50, 50, and 50) are all greater than 20 we do not have the sample distribution and cannot determine whether or not is is heavily skewed.

FREE-RESPONSE QUESTION 2

Calculate the value of the χ^2 statistic.

In addition to the original setup, use the following for Multiple Choice Questions 2 and 3: Suppose the value of the χ^2 statistic is 10.000 (which is not necessarily the correct answer to Free-Response Question 2 above!) Shown below are two potentially relevant figures:

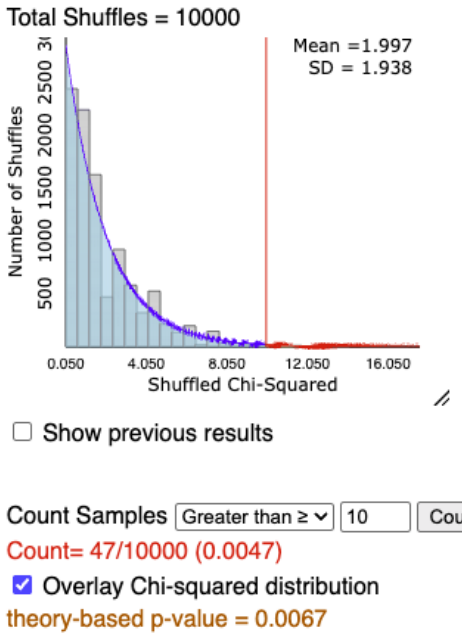


Figure 1: A screenshot from the Multiple Proportions applet

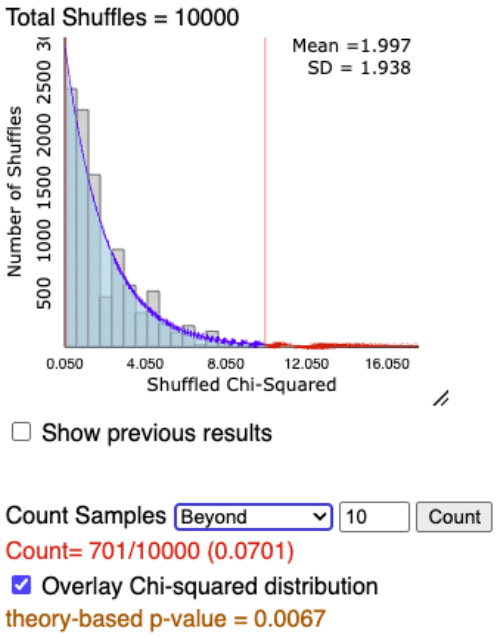


Figure 2: A screenshot from the Multiple Proportions applet

MULTIPLE CHOICE QUESTION 2

What is the value of the simulation-based *p*-value, using the χ^2 statistic?

(**A**) 0.0047 (**B**) 0.067 (**C**) 0.0701 (**D**) Something else

MULTIPLE CHOICE QUESTION 3

What is the value of the theory-based *p*-value, using the χ^2 statistic?

(**A**) 0.0047 (**B**) 0.067 (**C**) 0.0701 (**D**) Something else

FREE-RESPONSE QUESTION 3

Suppose the correct theory-based p-value is 0.01 (which is not necessarily the correct answer to Multiple Choice Question 3 above), and that the correct χ^2 statistic value is 10 (which is not necessarily the correct answer to Free-Response Question 2 above). Interpret the p-value.

Free-Response Question 4 is unrelated to the previous questions.

FREE-RESPONSE QUESTION 4

Explain why we should only look at the pairwise confidence intervals if we have rejected the null hypothesis of "no association." Make sure your explanation references the multiple testing problem. **Hint:** the chances of committing at least one Type I error among 3 hypothesis tests conducted on the same dataset are around 14.25%.

Some Potentially Useful Formulas and Definitions

$$z = \frac{(\hat{p}_1 - \hat{p}_2)}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}} \qquad t = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \qquad t = \frac{\bar{x}_d}{\left(\frac{s_d}{\sqrt{n}}\right)}$$

$$\text{Mean Group Diff} = \text{Mean of Absolute Values of Pairwise Differences in Sample Proportions} \qquad \chi^2 = \sum_{i=1}^{n} \left[\frac{(\widehat{p}_i - \widehat{p})}{\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n_i}}} \right]^2$$

p-value	Strength of Evidence
0.10 < p-value	Not much evidence against the null hypothesis; null is plausible
0.05 < p-value ≤ 0.10	Moderate evidence against the null hypothesis
0.01 < p-value ≤ 0.05	Strong evidence against the null hypothesis
p-value ≤ 0.01	Very strong evidence against the null hypothesis

Table 3: Guidelines for evaluating the strength of evidence from a p-value